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Enhanced Well Placement Through Infill Well Optimization with 3D Confidence Index Map

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Abstract

The success of a field development plan is highly dependent on the choice of well locations. Choosing the optimum well locations is, however, non-trivial and requires a thorough understanding of hydrocarbon reservoir performance and properties [1]. Even after the locations are identified there is a need to further rank the wells in terms of confidence as part of the risk assessment. This paper presents a novel approach that integrates geological modeling, geostatistics, and machine learning techniques to generate a confidence index map, which could then be used for optimizing infill well placement with a much higher level of confidence. This, in turn, helps with better decision making and reduces the risk involved in drilling additional infill wells.

The conventional approaches use averaging techniques to combine several reservoir attributes to identify the wells placement targets. These approaches could be further improved with the proposed workflow and capture the reservoir understanding which vary from one hydrocarbon reservoir to another. Leveraging on the various machine learning algorithms, the 3D Confidence Index Map (CIM) methodology addresses this challenge by systematically quantifying the uncertainty associated with key reservoir parameters and integrating them into a spatial map that highlights areas with the highest confidence for successful infill drilling.

This paper presents a workflow using the Confidence Index Map (CIM) methodology to addresses the challenge in optimizing infill well placement with a much higher level of confidence. Furthermore, this approach also provides valuable insights by incorporating uncertainty into the analysis, helps to identify and mitigate risks associated with infill drilling from mature oil and gas fields.

Introduction

The success of a field development plan in the oil and gas industry is heavily dependent on the strategic placement of wells. Identifying the optimal well locations is a complex task that requires a comprehensive understanding of the hydrocarbon reservoir's performance and properties. Even after potential locations have been identified, there is a need to further assess and rank these locations based on confidence levels as part of a thorough risk assessment. This process is crucial because drilling infill wells in mature fields can be a high-risk, high-reward endeavor. Infill wells are drilled to extract remaining hydrocarbons from fields that have been in production for some time, and their success can significantly impact the overall field development plan.

The conventional methods for optimizing infill well placement often rely on averaging techniques to combine various reservoir attributes. However, these approaches can be limited by their inability to fully capture the nuances and variability of different hydrocarbon reservoirs. Moreover, they may not adequately address the uncertainty associated with key reservoir parameters, which can lead to suboptimal well placement decisions. To overcome these challenges, there is a growing need for innovative methodologies that can integrate geological modeling, geostatistics, and machine learning techniques to provide a more accurate and confident approach to infill well optimization.

The 3D nature of the CIM will enable the visualization of the reservoir's properties and behavior in a more intuitive and comprehensive manner. This will allow for a better understanding of the relationships between different reservoir parameters and the identification of potential sweet spots that may have been overlooked using traditional 2D methods. The 3D CIM will also enable the evaluation of the reservoir's properties and behavior at different scales, from the individual well to the entire field, providing a more holistic understanding of the reservoir's performance.

The significance of this work lies in its ability to enhance well placement through infill well optimization with a higher level of confidence. It presents a workflow that can be applied to various hydrocarbon reservoirs, providing a tailored approach to each field's unique characteristics. The outcomes of this research have the potential to impact the oil and gas industry by improving the efficiency and effectiveness of field development plans, reducing the financial and operational risks associated with infill drilling, and contributing to the overall sustainability of hydrocarbon production. As the industry continues to evolve and face new challenges, innovative methodologies like the CIM will play a crucial role in optimizing resource extraction and ensuring the long-term viability of oil and gas operations.

Objectives/Scope

The objective of a Confidence Index Map is to provide a visual representation of confidence levels across different reservoirs attributes, properties and time periods. The application of the CIM methodology addresses a critical gap in current practices by providing a data-driven, spatially explicit approach to infill well placement. It not only helps in identifying "sweet spots" with favorable reservoir properties and high production potential but also offers valuable insights into the uncertainties associated with infill drilling. By incorporating uncertainty into the analysis, this approach enables the identification and mitigation of risks associated with infill drilling in mature oil and gas fields, ultimately leading to more informed decision-making and the potential to unlock the full potential of these fields.

This paper introduces a novel approach to infill well placement optimization through the development of a 3D Confidence Index Map (CIM). The CIM methodology leverages multiple layers of data, including pressure, saturation, porosity, permeability, history matching, and production data, to generate a comprehensive understanding of the reservoir. However, the main three attributes used in the confidence index are:

1. **History Matching** (Well Coverage): This attribute valuates the accuracy of the reservoir model by comparing predicted production data with actual production data. A history matching score of 0 indicates a perfect match, while a score of 3 indicates a significant deviation from the actual data.
2. **Sweet Spot** (Areal Coverage): This attribute identifies areas with the most favorable reservoir properties, such as high porosity and permeability, which are likely to produce hydrocarbons at a higher rate.
3. **Heterogeneity Index** (Vertical Zonation): This attribute assesses the variability of reservoir properties within each vertical zone, which can impact the flow of fluids and the overall production performance.

The BRUGGE model is used as the model of choice for this study, due to its ability to accurately capture the complex behavior of hydrocarbon reservoirs. By integrating these three maps, the CIM methodology provides a robust and comprehensive approach to infill well placement optimization. The resulting 3D Confidence Index Map will provide a detailed, spatially explicit representation of the reservoir, allowing for the identification of areas with the highest confidence for successful infill drilling.

Methods, Procedures, Process

Sweet Spot

Multiple reservoir variables were used to identify the sweet spots in the reservoir by following the steps below to result in ROI.

$$RQI = 0.0314\sqrt{\frac{k}{Ø}} \qquad \text{equation (1)}$$

Where:
RQI=Reservoir Quality Index
k=Permeability (mD)
Ø=Porosity (%)

The Reservoir Quality Index (RQI) is a critical parameter used in reservoir characterization to evaluate the quality of a reservoir by quantifying the variations in fluid flow throughout the reservoir. These variations are primarily influenced by the differences in lithologies, which are the physical properties and characteristics of the rock layers within the reservoir.

$$SOMPV = D_x \times D_y \times D_z \times Ø \times (S_o - S_{or}) \qquad \text{equation (2)}$$

Where:
SOMPV=Movable/Mobile Oil Volume
D_x, D_y, D_z=*Grid Simulation Dimentions*
Ø=Porosity (%)
S_o=*Oil Saturation at The Last Time Step*
S_{or}=*Irreducible Oil Saturation*

SOMPV shows the movable/mobile oil volume at the relevant time step. It is a vital parameter for assessing the producible volume of hydrocarbons in a reservoir, considering the dynamic nature of fluid flow and saturation changes over time

$$ROI = \sqrt{RQI \times SOMPV} \qquad \text{equation (3)}$$

Where:
ROI= Reservoir Opportunity Index (%)
RQI=Reservoir Quality Index (%)
SOMPV=Movable/Mobile Oil Volume

ROI represents the sweet spot percentage in each grid in the 3D reservoir map.

History Matching Quality Index

The History Matching Quality Index (HMQI) is a statistical measure used to quantify the quality of history matching between simulated (modeled) data and observed (historical) data in reservoir simulation studies. There is no standard approach to quantify the quality of history matching, however most common used are Normalized Root Mean Squares Error (NRMSE) as illustrate below:

Normalized Root Mean Squared Error (NRMSE):

$$NRMSE = \frac{\text{RMSE}}{\text{Range of Observed Data}} \qquad \text{equation (4)}$$

Where:

$$RMSE = \sqrt{\frac{1}{N}\sum\left(y_{obs} - y_{sim}\right)^2} \quad \text{equation (5)}$$

History Matching Quality Index (HMQI):

$$HMQI = 1 - \frac{NRMSE}{Var\left(y_{obs}\right)} \quad \text{equation (6)}$$

Where Var(yobs) is the variance of the observed data.

On this example, a higher HMQI value indicates a better match between the simulated and observed data and vice versa. HMQI value indicates a poorer match. Furthermore, the exact interpretation depends on the formulation of the HMQI, but it is often scaled between 0 and 1, where:

1: Perfect match (simulated data exactly reproduces observed data).

0: No correlation (simulated data does not match observed data at all).

Heterogeneity Index

While there are a few definitions for reservoir heterogeneity, in this paper, reservoir heterogeneity is considered to be the variations of reservoir rock properties in a particular zone, or reservoir.

In subsurface modeling, reservoir heterogeneity, while not explicitly evaluated in the entire 3D geological model, is very essential in order to appropriately evaluate how rock properties change within the geological model. It plays a major role in field development as it impacts fluid flow and overall reservoir performance, and if not accounted for, it could cause inefficient sweep, as well as uneven distribution of fluids.

Dykstra and Parsons [2] created a normalized scale that to measure heterogeneity as per Equation below:

$$Permeability\ Variation \quad V = \frac{stdev(\log k)}{avg(\log k)} = \frac{\log(k)_{P50} - \log(k)_{P84.1}}{\log(k)_{P50}} \quad \text{equation (7)}$$

Schmalz and Rahme in 1950 [3], came up with the Lorenz coefficient, which provides a single value to represent a reservoir's permeability distribution as a function of permeability, thickness, and porosity. The Lorenz coefficient is a normalized scale that describes a perfectly uniform homogenous reservoir with a value of 0, while a completely heterogeneous reservoir is represented by a value of 1 which reflects big variations in rock properties.

Figure 1 illustrates the Lorenz coefficient approach utilizing synthetic datasets. The black squares represent high reservoir property variations, while the blue circles show a more homogenous performance across the well.

Hamam and Maucec [4] expanded on the Lorenz approach to measure 3D heterogeneity index, which is one of the main indices to create the Confidence Index Map (CIM). To create the 3D heterogeneity index, the approach starts by data extraction

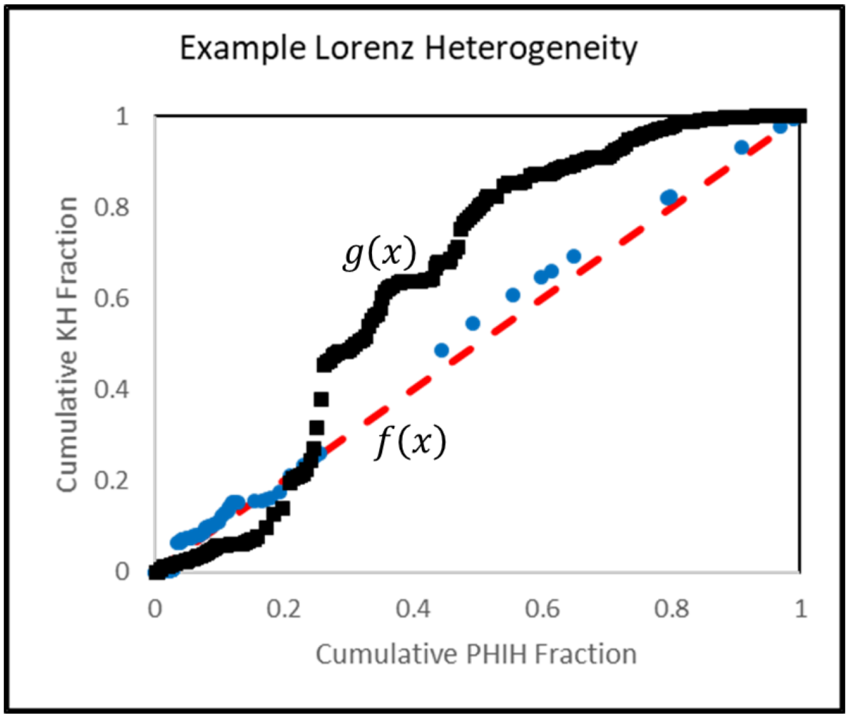

Figure 1—Cum PHIH Fraction vs. Cum KH Fraction (Lorenz Plot)

The process described by Hamam and Maucec [4] starts with data extraction from the geological model, and then vertical heterogeneity is calculated across all of the reservoir sections. Equations 8 through 15 shows the Lorenz coefficient calculations that are used to generate the heterogeneity index along a particular interval:

$$Storage\ Capacity\ (ØH) = Ø * \ Thickness \quad \text{equation (8)}$$

$$Flow\ Capacity\ (KH) = K * \ Thickness \quad \text{equation (9)}$$

$$Storage\ Capacity\ Fraction\ \ (ØHF) = \frac{Ø * \ Thickness}{Total\ ØH} \quad \text{equation (10)}$$

$$Flow\ Capacity\ Fraction\ (KHF) = \frac{K * \ Thickness}{Total\ KH} \quad \text{equation (11)}$$

$$Cumulative\ Storage\ Capacity\ Fraction\ (ØHF) = \sum_{n}^{1} ØHF \quad \text{equation (12)}$$

$$Cumulative\ Flow\ Capacity\ Fraction\ (KHF) = \sum_{1}^{n} KHF \quad \text{equation (13)}$$

Lorenz is the plot, however to calculate the heterogeneity index, it is the area bounded between the curve and the line through the origin, formula below:

$$Heterogeneity\ Index\ = \ Area\ Under\ Curve,\ g(x) - \ Area\ Under\ Line,\ f(x) \quad \text{equation (14)}$$

$$Heterogeneity\ _{Index} = \int_{0}^{1} [g(x) - f(x)]dx \quad \text{equation (15)}$$

Where g(x) = Area under the Lorenz curve from (0 – 1) from figure 1.
Where f(x) = Area under the Line through the Origin from (0 – 1) from figure 1.

After that, horizontal heterogeneity index is then calculated and mapped across the entire 3D model [Figure 2]. One major application of the 3D heterogeneity map is the utilization of the approach in well placement optimization. The 3D heterogeneity property is used to guide and to ensure that well placement sustains the same reservoir quality without abrupt change in rock properties across a particular reservoir zone or interval. However, certain features complicate the mapping, such as fractures, faults, super Ks, Tar, etc. and all of which are part of the heterogeneity methodology developed by Hamam and Marko [4].

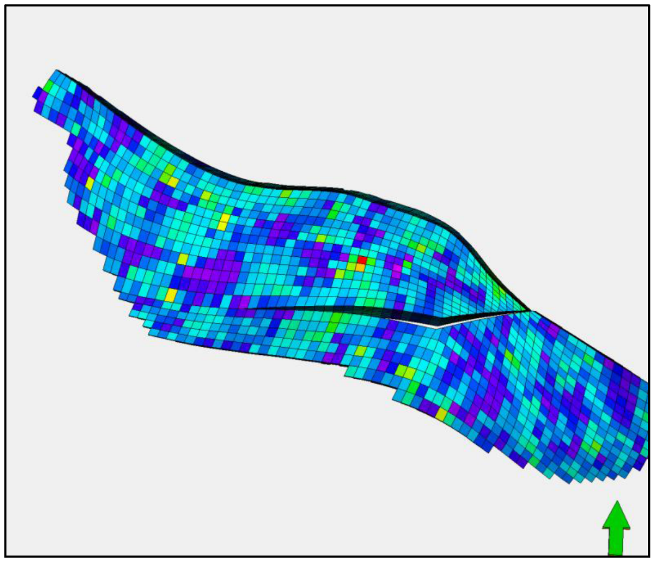

Figure 2—Final 3D Heterogeneity property mapped across the entire Zone/Reservoir

Model Description

BRUGGE benchmark reservoir model is used to demonstrate the effectiveness of the propose workflow. The structure of the Brugge field consists of an east/west elongated half-dome with a large boundary fault at its northern edge and one internal fault with a modest throw at an angle of approximately 20° to the boundary fault at the northern edge [5].

The full field model is about 60,000 Cartesian grid cells (139 × 48 x, consists of faulted blocks with 30 wells (20 producers and 10 injectors) as illustrated in the figure 3 below.

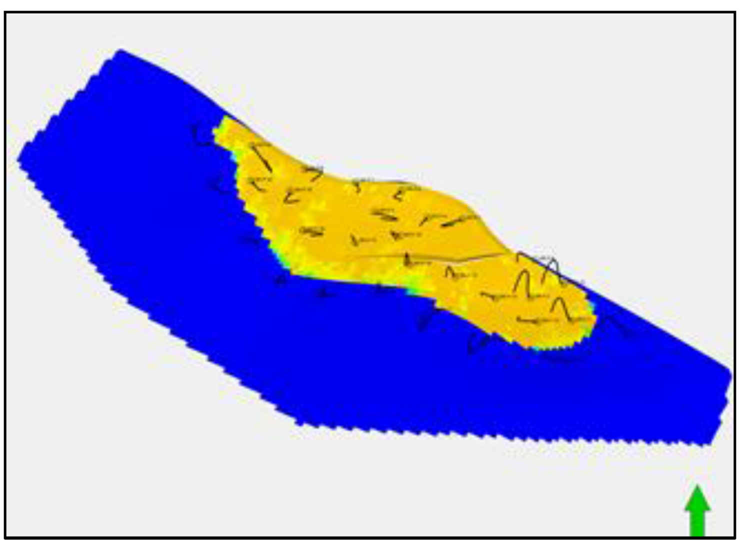

Figure 3—The 3D Full Field BRUGGE Simulation Model

For the purposes of this paper, the Porosity is generated stochastically using sequential Gaussian simulation and the permeability were generated deterministically as a single porosity/permeability regression with co-Kriging on porosity.

The average properties of the models as shown in the table below**:**

Table 1—The Average properties of 3D Full Field BRUGGE Simulation Model

Layers	Thickness, ft	$\emptyset_{avg}$, %	K_{avg}, md	Sw_{avg}, %
1	17.41	12.4	550.5	18.7
2	17.41	12.1	518.1	17.7
3	20.90	10.9	398.8	21.7
4	20.89	11.8	432.6	19.5
5	20.89	11.1	445.7	20.9
6	31.52	10.8	459.3	16.8
7	31.50	11.2	401.2	17.4
8	31.44	9.2	320.8	17.6
9	19.27	13.4	591.1	22.7

The model was initialized and history matched with the given observed data for both rate and pressure. The well history matching quality were shown in the figure 4 below which indicated the red color as poor history matching quality.

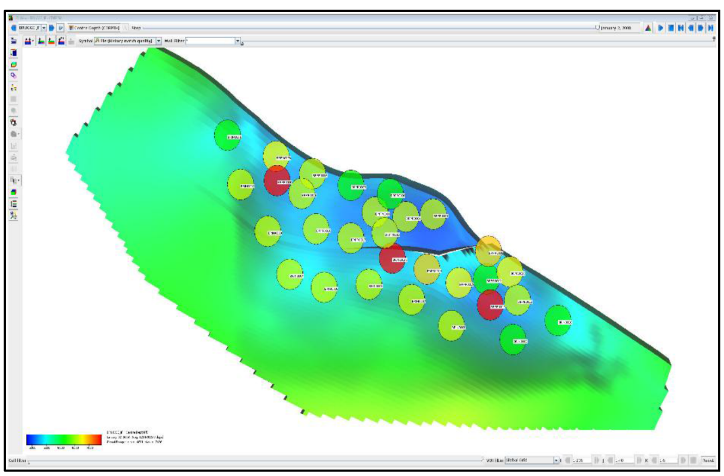

Figure 4—Well Level History Matching Quality

The quality assessment provides insights into the model's performance and guides further improvements. By addressing the areas of poor matching quality, we can enhance the model's predictive capabilities and ensure that it remains a valuable tool for decision-making.

Results and Observations

As discussed previously, the history matching quality were calculated for each well and mapped between the wells using simple kriging algorithm as shown in Figure 5a as 3D arrays grid. Furthermore, the heterogeneity index and reservoir opportunity index were calculated in 3D model as shown in the figure 5b and 5c.

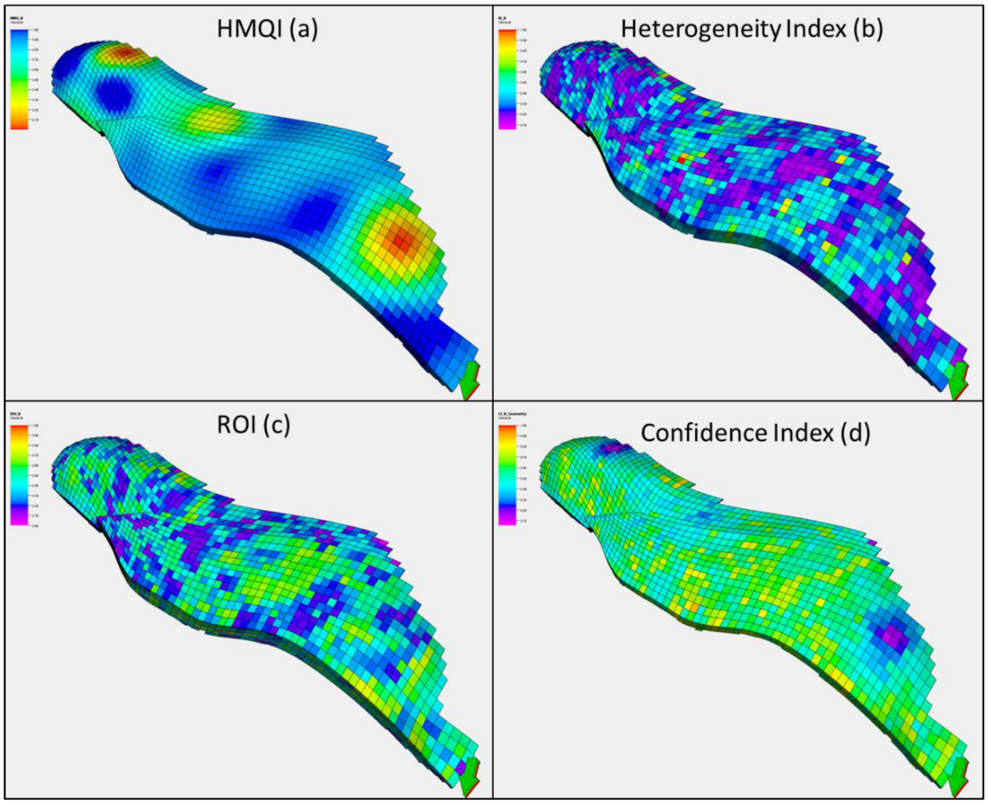

Figure 5—The 3D Array of HMQI, HI, ROI and CI

The Confidence Index 3D array then were generated based on the geometric average from these generated attributes as follow:

$$CIM = \sqrt[3]{HMQI * HI * ROI} \quad \text{equation (16)}$$

The Sweet Spot Map identifies areas with the most favorable reservoir properties, such as high porosity and permeability, which are likely to produce hydrocarbons at a higher rate. The History Match Quality index is a statistical measure of how the model is able to replicate observed (historical) data. The Heterogeneity index will assess the variability of reservoir properties within each vertical zone, which can impact the flow of fluids and the overall production performance.

By aggregating these maps as a result of the 3D Confidence Index Map (Figure 6) provides a detailed, spatially explicit representation of the reservoir, allowing for the identification of areas with the highest confidence for successful infill drilling. The 3D nature of the CIM enables the visualization of the reservoir's properties and behavior in a more intuitive and comprehensive manner.

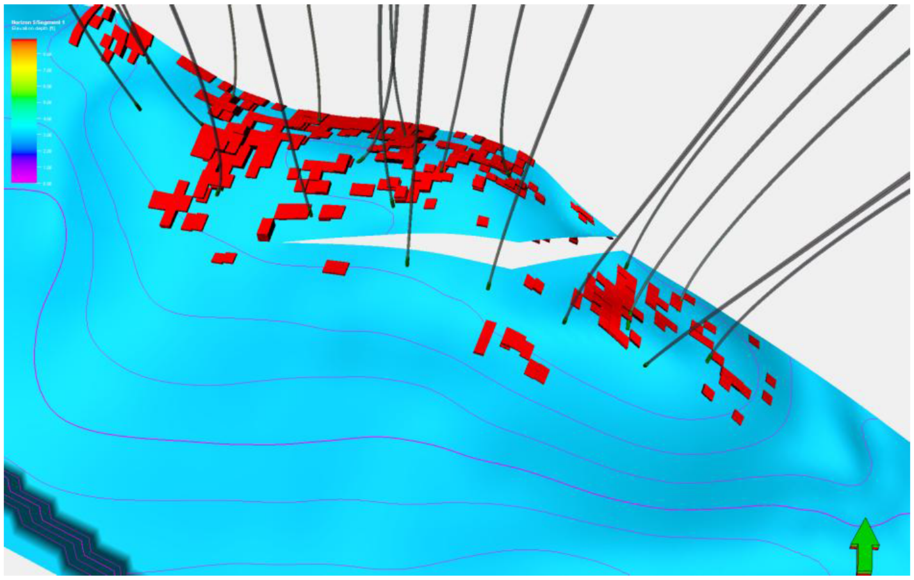

Figure 6—3D Confidence Index Map Showed Confidence Location to be Drilled

Conclusion

The development of a 3D Confidence Index Map (CIM) for optimizing infill well placement in mature oil and gas fields has been presented in this paper. The CIM methodology integrates multiple user-defined reservoir attributes layers of data, including saturation, porosity, permeability and history matching quality to generate a comprehensive understanding of the reservoir. The main three attributes used in the confidence index are the Sweet Spot, History Matching, and Heterogeneity Index.

The application of the 3D CIM methodology addresses a critical gap in current practices by providing a data-driven, spatially explicit approach to infill well placement. It not only helps in identifying "sweet spots" with favorable reservoir properties and high production potential but also offers valuable insights into the uncertainties associated with infill drilling. By incorporating uncertainty into the analysis, this approach enables the identification and mitigation of risks associated with infill drilling in mature oil and gas fields, ultimately leading to more informed decision-making and the potential to unlock the full potential of these fields.

The significance of this work lies in its ability to enhance well placement through infill well optimization with a higher level of confidence. It presents a workflow that can be applied to various hydrocarbon reservoirs, providing a tailored approach to each field's unique characteristics. The outcomes of this research have the potential to impact the oil and gas industry by improving the efficiency and effectiveness of field development plans, reducing the financial and operational risks associated with infill drilling, and contributing to the overall sustainability of hydrocarbon production.

In conclusion, the development of a 3D Confidence Index Map offers a novel and innovative approach to infill well placement optimization. The integration of the Sweet Spot, History Matching, and Heterogeneity Index provides a detailed understanding of the reservoir's properties and behavior, allowing for the identification of areas with the highest confidence for successful infill drilling. The 3D CIM methodology has the potential to significantly impact the oil and gas industry, improving the efficiency and effectiveness of field development plans and contributing to the overall sustainability of hydrocarbon production. As the industry continues to evolve and face new challenges, innovative methodologies like the 3D CIM will play a crucial role in optimizing resource extraction and ensuring the long-term viability of oil and gas operations.

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